REACTION IN BATCH REACTOR

Name: Purva Kaushalbhai Shah

Group members name:
Rutvi Shah
Ashwini Bukke
Kaila Uday
Md Sibtain Raza

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Objective:
The aim of this experiment is to determine the rate constant of a reversible second-order saponification reaction at room temperature using a batch reactor.

Theory and Background:

A batch reactor is a closed vessel commonly used in the process industry for conducting chemical reactions, especially for small-scale production and research purposes. In this type of reactor, all the reactants are added at the beginning, thoroughly mixed, and allowed to react over a fixed period. The contents are stirred continuously to maintain uniform conditions throughout, so every part of the mixture experiences the same reaction time.

The Saponification reaction of Sodium hydroxide with Ethyl Acetate is :

$$CH_3COOC_2H_5 \ + \ NaOH \ \rightarrow \ CH_3COONa \ + \ C_2H_5OH$$

This is a second-order reaction, which means the rate depends on the concentrations of both ethyl acetate and sodium hydroxide. Although the reaction is technically reversible, under room temperature conditions the equilibrium constant K_C is very big. This means the reaction proceeds almost entirely in one direction, allowing it to be treated as **irreversible** for practical purposes.

$$Rate \ = \ -\frac{d(NaOH)}{dt} \ = \ k\left([NaOH]\left[CH_3COOC_2H_5\right] - \frac{\left[C_2H_5OH\right]\left[CH_3COONa\right]}{K_C}\right)$$

$$Rate \ = \ -\frac{d(NaOH)}{dt} \ = \ k[NaOH]\left[CH_3COOC_2H_5\right]$$

If the initial concentrations of both reactants are equal, then the equation simplifies to:

$Rate \ = \ k[NaOH]^2$

$Since\ [NaOH]_0 \ = \ \left[CH_3COOC_2H_5\right]_0,\ where\ [NaOH]_0\ is\ the\ initial\ concentration\ of\ NaOH$

$\left[CH_3COOC_2H_5\right]_0\ is\ the\ initial\ concentration\ of CH_3COOC_2H_5$

$$[NaOH] = \left[CH_3COOC_2H_5\right] = [NaOH]\left(1 - X_A\right)$$

$Therefore\ we\ have$: $$-\frac{d(NaOH)}{dt} \ = \ k[NaOH]^2\left(1 - X_A\right)^2$$

$$-\frac{dX_A}{dt} \ = \ k[NaOH]_0\left(1 - X_A\right)^2$$

$$-\frac{\Delta X_A}{\Delta t} \ = \ k[NaOH]_0\left(1 - X_A\right)^2$$

Apparatus:

1. Magnetic stirrer

2. Thermometer
3. Conductivity meter
4. Glass beakers
5. Measuring cylinder
6. Pippets
7. Stop watch

Chemicals required:
1. NaOH
2. Ethyl Acetate
3. Distilled water

Procedure:
1. Prepare a 0.05 N stock solution of sodium hydroxide (NaOH).
2. Take known concentrations of NaOH from the stock solution and measure their corresponding conductivity (in mS/cm) using the conductivity meter.
3. Plot a calibration curve of conductivity vs. NaOH concentration.
4. Calculate the volumes of ethyl acetate and NaOH needed to maintain this volume while ensuring a 1:1 molar ratio.
5. Add the calculated volume of NaOH solution into the batch reactor.
6. Start the stirrer to ensure uniform mixing.
7. Switch on the conductivity meter.
8. Add the calculated volume of ethyl acetate into the reactor.
9. Start the stopwatch simultaneously with the ethyl acetate addition.
10. Record the conductivity readings at regular time intervals: 60 s, 120 s, 180 s, 300 s, 420 s, 600 s, 900 s, etc.
11. Continue recording until the conductivity reading stabilizes, indicating completion of the reaction.

Known parameters:
- $Initial\ Concentration\ of\ NaOH\ (C_{A0})\ =\ 0.05\ N$
- $Molar\ mass\ of\ NaOH\ =\ 40\frac{g}{mol}$
- $Molar\ Mass\ of\ CH:\ 88.\ 11\ g/mol$
- $Density\ of\ Ethyl\ Acetate\ (\rho):\ 0.\ 902\ \frac{g}{m^3}$

Formulae and Calculation:
- $N\ =\ \frac{Gram\ of\ solute}{equivalent\ weight \times Volume\ of\ Solution\ (in\ L)}$, where N is the Normality of the solution.

Readings:

TABLE 1: TABLE CONTAINING CONDUCTIVITY AND CONCENTRATION DATA OF NaOH

Conductivity (mS)	Concentration (M))
10.86	0.05
5.046	0.025
2.08	0.0125
1.297	0.00625
10.86	0.05

TABLE 2: TABLE CONTAINING EXPERIMENTAL DATA

Time (s)	**Conductivity(mS)**
60	7.358
120	6.8
180	6.393
240	6.082
300	5.836
360	5.649
420	5.482
480	5.346
540	5.23
600	5.134
660	5.049
720	4.975
780	4.905
840	4.849
900	4.8
960	4.751
1020	4.71
1080	4.672
1140	4.638

1200	4.606
1260	4.579
1320	4.553
1380	4.527
1440	4.511
1500	4.488
1560	4.472
1620	4.45
1680	4.433
1740	4.42
1800	4.407
1860	4.395
1920	4.382
1980	4.376
2040	4.359
2100	4.35
2160	4.339
2220	4.331

TABLE 3: TABLE CONTAINING CALCULATED DATA

Time (s)	**Conductivity (mS)**	**C_A, conc**	**XA=1-CA/CA0**	**dXA/dt**	**CA0(1-xA)^2**
60	7.358	0.034741	0.305179	0.000830	0.024139
120	6.8	0.032250	0.355000	0.000606	0.020801
180	6.393	0.030433	0.391339	0.000463	0.018523
240	6.082	0.029045	0.419107	0.000366	0.016872
300	5.836	0.027946	0.441071	0.000278	0.015620
360	5.649	0.027112	0.457768	0.000249	0.014701
420	5.482	0.026366	0.472679	0.000202	0.013903
480	5.346	0.025759	0.484821	0.000173	0.013270

540	5.23	0.025241	0.495179	0.000143	0.012742
600	5.134	0.024813	0.503750	0.000126	0.012313
660	5.049	0.024433	0.511339	0.000110	0.011939
720	4.975	0.024103	0.517946	0.000104	0.011619
780	4.905	0.023790	0.524196	0.000083	0.011319
840	4.849	0.023540	0.529196	0.000073	0.011083
900	4.8	0.023321	0.533571	0.000073	0.010878
960	4.751	0.023103	0.537946	0.000061	0.010675
1020	4.71	0.022920	0.541607	0.000057	0.010506
1080	4.672	0.022750	0.545000	0.000051	0.010351
1140	4.638	0.022598	0.548036	0.000048	0.010214
1200	4.606	0.022455	0.550893	0.000040	0.010085
1260	4.579	0.022335	0.553304	0.000039	0.009977
1320	4.553	0.022219	0.555625	0.000039	0.009873
1380	4.527	0.022103	0.557946	0.000024	0.009771
1440	4.511	0.022031	0.559375	0.000034	0.009708
1500	4.488	0.021929	0.561429	0.000024	0.009617
1560	4.472	0.021857	0.562857	0.000033	0.009555
1620	4.45	0.021759	0.564821	0.000025	0.009469
1680	4.433	0.021683	0.566339	0.000019	0.009403
1740	4.42	0.021625	0.567500	0.000019	0.009353
1800	4.407	0.021567	0.568661	0.000018	0.009303
1860	4.395	0.021513	0.569732	0.000019	0.009257
1920	4.382	0.021455	0.570893	0.000009	0.009207
1980	4.376	0.021429	0.571429	0.000025	0.009184
2040	4.359	0.021353	0.572946	0.000013	0.009119
2100	4.35	0.021313	0.573750	0.000016	0.009084
2160	4.339	0.021263	0.574732	0.000012	0.009043
2220	4.331	0.021228	0.575446		0.009012

Results and discussion:

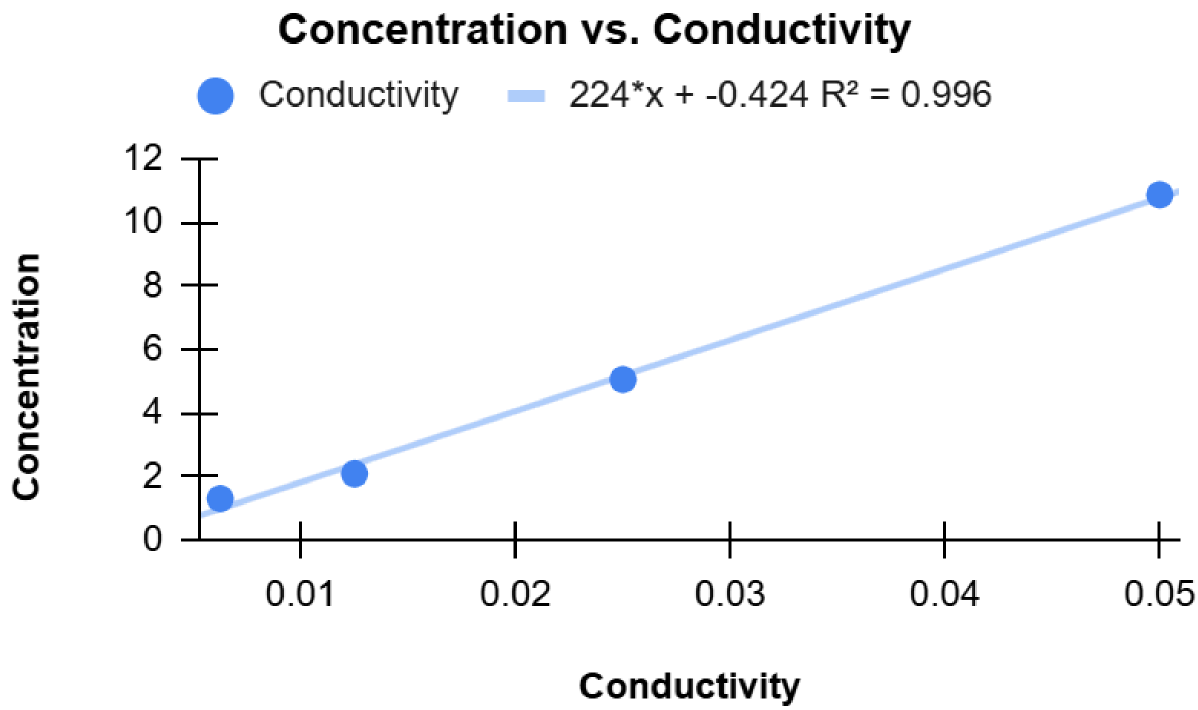

Figure 1: Figure showing the Concentration (M) vs Conductivity (mS)curve (Calibration curve)

From the plot we get, $Conductivity\ = \frac{concentration + 0.424}{224}$

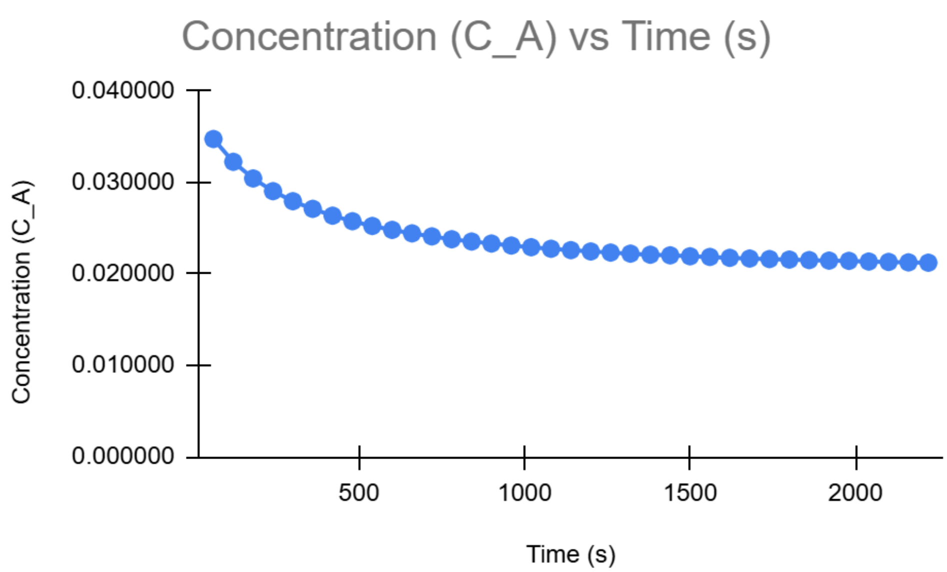

Figure 2: Figure showing the plot between Concentration (M) and Time (s)

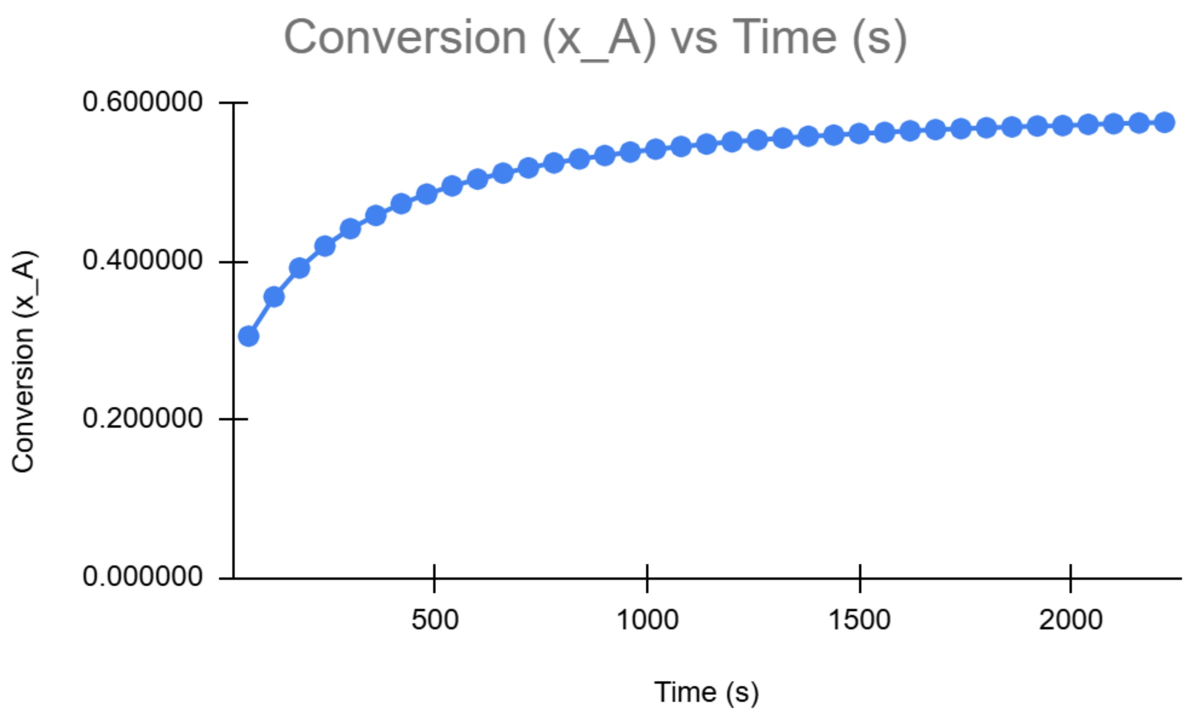

Figure 3: Figure showing the plot between Conversion (X_A) and Time (s)

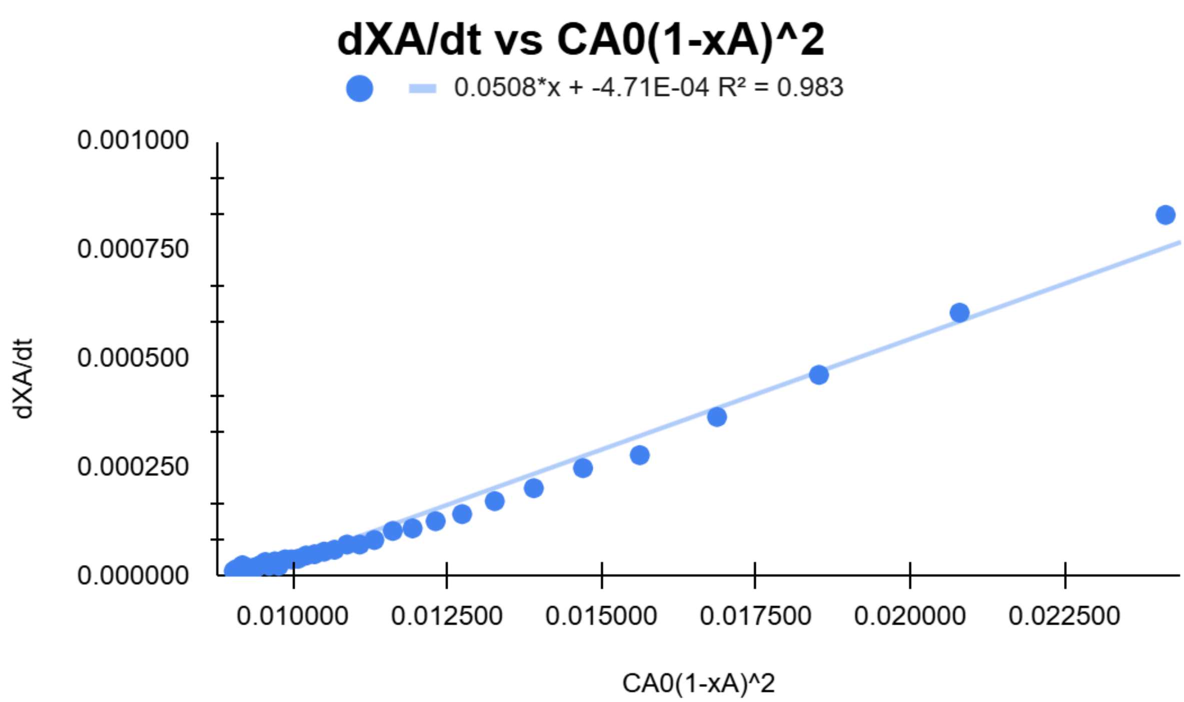

Figure 4: Figure showing the plot between $\frac{dX_A}{dt}$ $vs\ C_{A0}(1-X_A)^2$

Figure 1 shows the relationship between the conductivity of NaOH solutions and their corresponding concentrations. A strong linear trend was observed ($R^2 = 0.996$), confirming that conductivity can be reliably used to determine NaOH concentration. The calibration curve equation derived from this plot was later used to convert real-time conductivity measurements into accurate concentration values during the reaction.

Figure 2 shows how the concentration of NaOH changes over time during the saponification reaction. Initially, the conductivity and hence concentration was high due to the presence of ionized NaOH. As the reaction progressed, NaOH was gradually consumed, forming sodium acetate and ethanol, which are less conductive.The curve exhibits a nonlinear decline, which is typical of second-order kinetics, where the rate depends on the product of reactant concentrations.The NaOH concentration started at 0.05 mol/L and decreased significantly, particularly in the first few minutes, as the reaction proceeded rapidly.

Figure 3 shows the conversion of NaOH over time. As expected, the conversion increases steadily, reflecting the continuous consumption of NaOH as it reacts with ethyl acetate. Initially, the rate of conversion is high due to a higher concentration of reactants. Over time, the curve levels off, indicating the system is approaching equilibrium. The final conversion achieved was approximately 0.57, meaning more than half of the initial NaOH was used up in the reaction.

Figure 4 analyzes the rate behavior of the saponification reaction and validates the second-order kinetics. As the reaction progressed, the rate of reaction $\frac{-dC_A}{dt}$ gradually decreased due to the diminishing concentrations of the reactants. A plot $\frac{dX_A}{dt}$ $vs\ C_{A0}(1 - X_A)^2$ produced a **linear trend**, which confirmed that the reaction follows a **second-order rate law**. The rate constant for the saponification of ethyl acetate with NaOH at room temperature is $0.0508\ \frac{L}{g\ mol\ sec}$

Conclusion:

The experiment successfully demonstrated the second-order kinetics of the saponification reaction between ethyl acetate and sodium hydroxide in a batch reactor. Initially, the solution exhibited high conductivity due to the presence of ionized NaOH. As the reaction proceeded, conductivity decreased steadily with the formation of sodium acetate and ethanol, which are less conductive.

The calibration curve enabled accurate conversion of conductivity data into NaOH concentrations. The nonlinear decline in concentration over time, along with a linear fit in the rate analysis plot, validated the second-order nature of the reaction. The reaction rate constant (k) was determined to be $0.0508\ \frac{L}{g\ mol\ sec}$.

Conversion of NaOH increased with time and gradually approached equilibrium. The maximum conversion observed was approximately **0.57**, indicating that over half of the initial NaOH had reacted.

Despite minor experimental limitations, including possible conductivity meter sensitivity and minor temperature fluctuations, the data was consistent and aligned with theoretical expectations. This highlights the effectiveness of batch reactors in small-scale kinetic studies and reinforces the importance of precision in calibration and measurement.

Sources of errors:

- Slight inaccuracies in conductivity readings due to limited sensitivity of the instrument.
- Minor temperature variations during the experiment that may have influenced the reaction rate.
- Inconsistent stirring possibly causing uneven mixing of reactants.
- Human error in noting time stamps or recording conductivity values.
- Delays or mismatches during reagent addition and stopwatch activation.

Industrial Applications:

- It is used in industries producing soaps, detergents, and biodiesel, where saponification reactions between esters and strong bases like NaOH are common.
- It is used in pharmaceutical industries where small volumes of high-value products are synthesized with strict control over reaction time and conditions.
- It is used in specialty chemical production where flexibility and the ability to safely handle toxic or highly potent substances are required.
  It is used in food and fragrance industries for ester hydrolysis processes where precise conversion tracking is essential for product consistency.

Academic relevance:

- It is used for studying reaction kinetics, including the determination of rate constants, activation energy, and reaction order under controlled lab conditions.
- It is ideal for academic research where small-scale batch setups allow precise observation of concentration changes and conversion over time.
- It is used in university labs to demonstrate fundamental principles of chemical kinetics and reaction engineering to students.
- It is used in research settings to model complex reactions using simpler systems like the saponification of ethyl acetate and NaOH.

References:

[1]. O. Levenspiel, Chemical Reaction Engineering, 3rd ed. Wiley, 1999.
[2] ‘Lab Manual’ provided in the classroom.